

Robotics Dynamics Under a New “Lens”

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List of concepts being used

- Hyper Dual numbers and Clifford’s Geometry
- Principle of least action
- General Relativity and proper time
- Fermat's Principle of Least Time
- Calculus of Variations
- Controls Theory

Introduction

"If you wish to make an apple pie from scratch, you must first invent the universe." — Carl Sagan

"What is the Universe?" is an unconventional question to open a paper on robotics, yet establishing this definition is critical. At its most fundamental level, the universe can be modeled as a machine that evolves states into other states through the expenditure of time. This local, step-by-step evolution lies at the heart of cellular automata like Conway’s *Game of Life*—a zero-player system where a simple seed and a set of local instructions cause a complex "universe" to unfold. Conceptually, it is an initial value problem governed by constraints that dictate its continuous evolution. In robotics, these constraints are mathematically formulated as the laws of physics, most commonly through Newtonian or Lagrangian mechanics. While Newtonian mechanics studies bodies in motion under the direct influence of vector forces, Lagrangian mechanics proposes a fascinating alternative: that the universe operates by minimizing a conceptual currency called "Action" as a conceptual currency, however, has historically been a source of philosophical friction; it lacks intuitive meaning, particularly at a macro scale. Yet, its validity is inarguable. It mathematically forms the foundation of modern robotics. The issue with Action is not a lack of mathematical rigor or empirical testing, but rather a fundamental disconnect regarding its physical significance. The goal of this report is to demonstrate that just as Newton’s laws emerge from the minimization of Action, Action itself is a byproduct of a more primordial principle. If the universe is a machine that evolves states into other states by spending time, it stands to reason that time itself should be the fundamental currency.

What is “Action” ?

The historical development action is a fascinating bit of progression in the tapestry of physics. However to explain how it came to be we have establish what came before, primarily “Fermat’s Principle of Least Time”, which describes an optic phenomenon called refraction. Fermat illustrated that the light took the path that minimized the time it takes, as a result refracting. This was done by defining the total time in terms of time spent in each medium, which was equal to the distance traveled in that medium divided by the speed of light in that medium. When the times are plotted against the possible points of refractions, denoted as x , a concave up function is created, meaning when the functions gradient is equal to zero, the minimum is reached. Which just so happens to align with the real world point of refraction, meaning light took the path that minimized its own time. Fermat later wrote that "The result of my computation was the most extraordinary, the most unexpected, and the happiest that could ever be. [...] The price of my triumph is the most beautiful thing in the world." . This was the first inkling in physics that pointed to the possibility of the universe trying to optimize for something.

Normally the field of “Optics” and “Mechanics” are totally separate fields that operate under different conventions and notations, however a significant intersection happened when trying to solve the “Brachistochrone problem” (Greek for “Fastest Descent”) where given a set of points what path results in a ball sliding down on it to. Johann Bernoulli recognized that the “Principle of Least Time” could be used to solve this problem if the our particle “refracted” continuously, it would follow the path that minimized its own time. This is one of the first concrete examples mechanics being described as a product of a minimization problem.

Following Bernoulli, Pierre Louis Maupertuis extended this minimization concept into a cosmic principle, defining a fundamental currency of the universe: Action. He formulated action as a discrete summation of momentum multiplied by distance:

$$Action = \sum m_i v_i s_i$$

Maupertuis' formulation drew immediate scrutiny. It lacked mathematical rigor, its physical interpretation was completely non-intuitive, and it was overtly teleological. Maupertuis went as far as arguing that the universe's efficiency was a definitive proof of God. However there were proponents of it as well, specifically Leonard Euler who later improved upon the mathematical rigor by defining several key concepts. Among which, was to change the discrete sum into a continuous integral along a spatial path.

$$Action = S = \int m v ds = \int m v^2 dt \int 2T dt$$

Importantly, Euler established that this formulation explicitly requires the total energy $E = T + U$ to be strictly conserved along the path.

However there were still problems with this way of working with the action, specifically that it was difficult to keep track of things like kinetic energy and velocity at every time step and to do the necessary calculations. However another key contribution was made to it by Joseph Louis Lagrange. Lagrange used the calculus of variations to streamline the process and actually made it a worthy alternative to Newtonian Mechanics, the resulting formulations layed the foundation of what would become Lagrangian Mechanics.

Calculus of Variations proof of Principle of Least Action

Firstly we define an arbitrary, continuous and differentiable function “ η ”. This, in conjunction with “ ϵ ” which represent an infinitesimal number that scales η , acts as tiny perturbations that we inject into a function, when we take the difference of the perturbed function with the regular functions it serves as almost like partial derivative that acts along the entire function, so rather than setting the partial derivatives of a point to zero we are able to set the entire functions partial derivative to be 0, to the first order at least

$$\eta : [t_0, t_f] \rightarrow \mathbb{R}^n, \quad \text{where} \quad \eta(t_0) = \eta(t_f) = 0$$

To aid in derivation it is helpful that we formulate Euler’s notation in an alternative way, where rather than just looking at $2T$ we look at the sum of the “Total Energy”, the “Kinetic Energy” and the minus “Potential Energy. For shorthand it is common to shorten $T-U$ to L , where L represents the Lagrangian, coined after Lagrange

$$S(q, \dot{q}) = \int 2T * dt = \int (Edt + \int (T - U)dt$$

Because we mainly concern ourselves with first order accuracy, and the fact that perturbations themselves are arbitrarily small we use the first order Taylor expansions to define the perturbed states denoted as “*”

$$T(q, \dot{q}) \rightarrow T^* = T(q + \epsilon\eta, \dot{q} + \epsilon\dot{\eta}) \approx T(q, \dot{q}) + \epsilon * (\eta * \frac{\partial T}{\partial q} + \dot{\eta} * \frac{\partial T}{\partial \dot{q}})$$

$$U(q) \rightarrow U^* = U(q + \epsilon\eta) \approx U(q) + \epsilon\eta * \frac{\partial U}{\partial q}$$

$$E(q, \dot{q}) = E \rightarrow E(q + \epsilon\eta, \dot{q} + \epsilon\dot{\eta}) = E$$

After defining the perturbed energies we use them in place of the unperturbed energies in the action formulation to get our perturbed action

$$\begin{aligned} S^* &= S(q + \epsilon\eta, \dot{q} + \epsilon\dot{\eta}) = \int Edt + \int (T(q, \dot{q}) + \epsilon * (\eta * \frac{\partial T}{\partial q} + \dot{\eta} * \frac{\partial T}{\partial \dot{q}})) - U(q) + \epsilon\eta * \frac{\partial U}{\partial q} dt \\ &= S + \epsilon \int (\eta * \frac{\partial T}{\partial q} + \dot{\eta} * \frac{\partial T}{\partial \dot{q}} - \eta * \frac{\partial U}{\partial q}) dt = S + \epsilon (\int (\eta * \frac{\partial T}{\partial q} - \eta * \frac{\partial U}{\partial q}) dt + \int \eta \frac{d}{dt} \frac{\partial T}{\partial \dot{q}} dt) \end{aligned}$$

By taking the difference between the perturbed and unperturbed function we are able to find where the action is stagnant.

$$S^* - S = \epsilon (\int (\eta * \frac{\partial T}{\partial q} - \eta * \frac{\partial U}{\partial q} - \eta \frac{d}{dt} \frac{\partial T}{\partial \dot{q}}) dt + \int \eta \frac{\partial T}{\partial \dot{q}}) = \epsilon (\int (\eta * \frac{\partial T}{\partial q} - \eta * \frac{\partial U}{\partial q} - \eta \frac{d}{dt} \frac{\partial T}{\partial \dot{q}}) dt) = 0$$

And because η is an arbitrary function, to ensure the integral always equals a zero we have to consider that all the other terms must equal zero.

$$\frac{\partial T}{\partial q} - \frac{\partial U}{\partial q} - \frac{d}{dt} \frac{\partial T}{\partial \dot{q}} = 0, \quad U = mgy, \quad T = \frac{1}{2}mv^2, \quad -\frac{\partial U}{\partial q} = F, \quad \frac{d}{dt} \frac{\partial T}{\partial \dot{q}} = \frac{d}{dt}mv = ma \rightarrow F = ma$$

Hyper Dual Numbers and Automatic Differentiation:

A “Dual Number” is a number that satisfies some unique conditions which allow for us to take derivatives without having to evaluate limits or finite differences. The imaginary number in question is “ ϵ ” and is defined as:

$$\epsilon^2 = 0 \text{ and } \epsilon \neq 0$$

Though at a first glance this seems arbitrary to define, when plugged into a function and then expanding said function using a Taylor series of a function centered around x , we get the following results

$$f(x + h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x) * h^n}{n!} \rightarrow f(x + \epsilon h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x) * (\epsilon h)^n}{n!} = f(x) + \epsilon * h * f'(x) + \epsilon^2 * h^2 * \frac{f''(x)}{2} \dots$$

The fact that all the terms of epsilon which have a power higher than 1 are equal to zero, this leaves us the expression:

$$f(x + \epsilon h) = f(x) + \epsilon h f'(x)$$

Which also means, that without evaluating limits or finite differences we are able to represent the derivative as the following relation:

$$\frac{\partial f(x + \epsilon h)}{\partial \epsilon} = f'(x) * h$$

This is the basis behind how automatic differentiation works where by interjecting this imaginary number we are able to compute derivatives without having to resort to finite differences. Hyper duals are also used for a similar purpose:

$$\epsilon_1^2 = 0, \epsilon_2^2 = 0, \epsilon_1 * \epsilon_2 = \epsilon_{12} \neq 0$$

$$f(x + \epsilon_1 h_1, y + \epsilon_2 h_2) = f(x, y + \epsilon_2 h_2) + \epsilon_1 h_1 * \frac{\partial f(x, y + \epsilon_2 h_2)}{\partial x} = f(x, y) + \epsilon_1 h_1 \frac{\partial f(x, y)}{\partial x} + \epsilon_2 h_2 \frac{\partial f(x, y)}{\partial y} + \epsilon_{12} * \frac{\partial^2 f(x, y)}{\partial x \partial y}$$

As can be seen hyper dual numbers allow us to take partial derivatives without having to evaluate limits or take finite differences. This is great in terms of computing, also it gives us a way to describe partial derivatives of a function as purely the imaginary part of the function, meaning the function itself carries its own derivatives inside it.

Principle of stagnant proper time

The results we get from the principle of least action are irrefutable, however strictly speaking this shouldn't distract from the fact that “action” is still a very unintuitive currency. It still has teleological properties meaning it requires things to peek into the future to decide how to behave in the present. This “out of time” aspect of action clashes with how the universe works, at its core the universe is a machine that evolves states into other states through time, but action seems to describe a system that devolves present states into future states which opposes the flow of time as we perceive it. This was the key concept that got me thinking about whether time was the quantity that was stagnated, rather than the action. However to explore this notion we have to consider what time is. In General relativity there is no time, nor space, there is spacetime. Spacetime describes a geometry where time and space coexist as related parameters and they are represented in Minkowski space where the magnitude is the Reimannian distance, the magnitude is referred to as the “proper time” which is represented by “ τ ”. In General Relativity things move along geodesics which are points where $\partial \tau = 0$. This is suspiciously similar to how we define a stagnant action, so in effect rather than the universe stagnating its action, it is stagnating its proper time which refers to just taking the straightest path that can be taken in the geometry. This achieves 3 crucial things. It rescues us from teleology, poses everything in terms of geometry, and actually provides us with an intuitive unit to work with.

Derivation of the fundamental equations of mechanics using principle of stagnant proper time:

In general relativity everything traces the path that maximizes its aging, the measure of this aging is defined in terms of the proper time .

$$\tau = \sqrt{t^2 - \frac{r^2}{c^2}}$$

Since general relativity's core postulate is that spacetime is continuous we are able to work with a differential piece of space time.

$$d\tau = \sqrt{(1 - \frac{v^2}{c^2})} dt$$

Another core concept borrowed over from general relativity is the definition of total energy where on top of the classical potential and kinetic energies we also add the base energy governed by the famous $e = mc^2$ equation

$$E_{total} = mc^2 + T + U(q) \rightarrow mc^2 = E_{total} - T - U$$

By expanding the differential proper time equation we are able to derive the definition of the squared differential proper time in terms of energies and time

$$d\tau^2 * c^2 = c^2 * dt^2 - v^2 * dt^2 \rightarrow mc^2 * d\tau^2 = mc^2 * dt^2 * mv^2 dt^2$$

$$(E_{total} - T - U) * d\tau^2 = (E_{total} - T - U) * dt^2 - 2 * T * dt^2 \rightarrow d\tau^2 = (1 - \frac{2T}{E_{total} - T - U}) dt^2$$

To get the equation for the actual differential proper time we need to take the square root, however we are able to use a first order Taylor expansion as the term inside the square root is incredibly close to 1 due to the sheer magnitude of the squared speed of light has, leaving us with:

$$d\tau = \sqrt{1 - \frac{2T}{E_{base}}} * dt = \sqrt{1 - \frac{E + L}{E_{base}}} * dt \approx (1 - \frac{E + L}{2 * E_{base}}) dt \text{ where } E = T + U + \int W_{nc} * dt$$

Instead of using the standard calculus of variations or Christofels symbols to define the geodesic we can exploit hyperdual numbers to streamline the process of deriving the equations of our perturbed states.

$$\epsilon_1^2 = 0, \epsilon_2^2 = 0, \epsilon_1 * \epsilon_2 = \epsilon_{12} \neq 0$$

$$dt^* = dt + \epsilon_1 dt, q^* = q + \dot{q} * dt^* + \epsilon_2 * \eta_q, dq^* = dq + \ddot{q} * (dt^*)^2 + \epsilon_2 \dot{\eta}_q * dt^*$$

$$d\tau^* = \left(1 - \frac{E_0 + T(\dot{\mathbf{q}}^*) - U(\mathbf{q}^*) + (\dot{\mathbf{q}}^{*T} \mathbf{Q}_{nc}) dt^*}{2E_{base}} \right) dt^*$$

$$T^* = T(\dot{q}) + \frac{\partial T}{\partial \dot{q}} (\ddot{q} dt - \epsilon_1 \dot{q} + \epsilon_1 \ddot{q} dt + \epsilon_2 \dot{\eta}_q) + \frac{1}{2} \frac{\partial^2 T}{\partial \dot{q}^2} ((\ddot{q} dt)^2 + \epsilon_1 [2(\ddot{q} dt)^2 - 2\dot{q} \ddot{q} dt] + \epsilon_2 [2\dot{\eta}_q \ddot{q} dt] + \epsilon_{12} [2\dot{\eta}_q \ddot{q} dt - 2\dot{q} \dot{\eta}_q])$$

$$U^* = U(q) + \frac{\partial U}{\partial q} (\dot{q} dt + \epsilon_1 \dot{q} dt + \epsilon_2 \eta_q) + \frac{1}{2} \frac{\partial^2 U}{\partial q^2} ((\dot{q} dt)^2 + 2\epsilon_1 (\dot{q} dt)^2 + 2\epsilon_2 \dot{q} \eta_q dt + 2\epsilon_{12} \dot{q} \eta_q dt)$$

By fully expanding we arrive at the very same Euler Lagrange formula as well as the constrained robot equation. All derived from stagnating proper time along a geometry.

$$\frac{\partial(d\tau^*)}{\partial \epsilon_2} = -\frac{1}{2E_{base}} \left[(\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})) \dot{\eta}_q - \frac{\partial U}{\partial \mathbf{q}} \eta_q + \mathbf{Q}_{nc}^T \dot{\eta}_q dt \right] dt$$

$$\frac{\partial^2(d\tau^*)}{\partial \epsilon_{12}} = -\frac{1}{2E_{base}} \left[\left(\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T}{\partial \mathbf{q}} \right) \eta_q - \frac{\partial U}{\partial \mathbf{q}} \eta_q \right] dt$$

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \frac{\partial U}{\partial \mathbf{q}} = \mathbf{Q}_{nc}$$